Does vibrational strong coupling reorganize the hydrogen-bond network of liquid water? Polarizable machine-learning cavity molecular dynamics

Charlie Wade[1] and Johannes Flick[1, 2, 3, *]

[1] *Department of Physics, City College of New York, New York, New York 10031, USA*
[2] *Department of Physics, The Graduate Center, City University of New York, New York, New York 10016, USA*
[3] *Center for Computational Quantum Physics, Flatiron Institute, New York, New York 10010, USA*

(Dated: November 11, 2026)

Infrared experiments on liquid water in Fabry–Pérot cavities indicate that vibrational ultra-strong coupling of the O–H stretch slows the thermal disordering of the hydrogen-bond network and redistributes water between network, intermediate and isolated configurations, without any external driving. A microscopic explanation is missing: in classical cavity molecular dynamics with fixed-charge models, the cavity cannot change equilibrium structure at all, so such simulations are blind to this effect by construction. Here we use `cboamd`, a cavity Born–Oppenheimer molecular-dynamics framework that includes the electronic polarizability, the minimal ingredient through which a cavity can bias thermal equilibrium. We validate the approach on single CO_2 and water molecules, follow the collective coupling from one to a few water molecules, and simulate liquid water with machine-learned models for its energy, dipole moment and polarizability, to ask whether, and how, the cavity reorganizes the hydrogen-bond network.

I. INTRODUCTION

When molecular vibrations interact strongly with the confined electromagnetic field of an optical cavity, they hybridize with the cavity mode into vibrational polaritons. In this regime of vibrational strong coupling (VSC), the infrared (IR) absorption peak of the molecules splits into a lower and an upper polariton separated by the Rabi splitting, which grows with the square root of the number of coupled molecules [1, 2]. What makes VSC remarkable is that it appears to change matter itself: placing reacting solutions in a resonant infrared cavity, in the dark, has been reported to slow reactions, change their selectivity and modify energy transfer [2–5]. At the same time, careful attempts to reproduce some of these effects found much smaller changes or none at all [6, 7]. Why, and when, a cavity alters the ground state of a molecular ensemble remains an open question.

Water is the natural place to ask this question. It is the solvent of most of chemistry and biology, and its properties are set by a flexible hydrogen-bond network. Its O–H stretch is one of the strongest infrared absorbers, so liquid water reaches the strong- and even ultra-strong-coupling regimes readily. Experiments have reported that collective VSC of water raises its ionic conductivity [8]. Most recently, Itatani *et al.* resolved the O–H stretching band of water under vibrational ultra-strong coupling into contributions from network, intermediate and isolated molecules [9]. They found that the thermal disordering of the hydrogen-bond network is retarded inside the cavity, that the fractions of intermediate and isolated water grow anomalously at elevated temperature, and that the upper polariton narrows towards the homogeneous linewidth of single water molecules. They interpret these observations as vacuum-field-mediated correlations between water molecules that extend beyond those of bulk water. A spectrum alone, however, cannot say which hydrogen bonds change, or why.

Molecular dynamics (MD) simulations can, because they follow every molecule. Cavity MD with empirical force fields has been applied to liquid water under VSC and reproduces the polariton splitting of the O–H stretch [10, 11], as does an extended cavity Born–Oppenheimer treatment [12]. Where the structure of the liquid was examined, it changed little [10, 12]. This is not an accident. For a fixed-charge model, the photon coordinate is a harmonic oscillator whose center depends on the nuclear positions but whose curvature does not; integrating it out of the classical partition function gives a constant, so every equilibrium average of the nuclei is identical inside and outside the cavity [13]. A classical simulation can only see a cavity-induced change of the hydrogen-bond network if the coupling depends on the configuration in a way that goes beyond a fixed dipole. In addition, fixed-charge models misrepresent the light–matter coupling itself: an inaccurate dipole surface directly distorts the Rabi splitting, and for water the widely used q-TIP4P/F model underestimates the O–H stretch intensity and therefore its splitting [11].

The cavity Born–Oppenheimer approximation (CBOA) [14] provides a first-principles route that avoids both problems. In its effective electric-field formulation [15], the coupled nuclear–photon dynamics depend only on three field-free properties of the matter: the ground-state energy, the dipole moment and the polarizability. All three can be computed with standard electronic-structure codes and learned by machine-learning interatomic potentials (MLIPs). The resulting framework, `cboamd` [16], makes collective VSC in the condensed phase accessible at near first-principles accuracy. Crucially, the polarizability screens the cavity field in a configuration-dependent way. This renormalizes the

* jflick@ccny.cuny.edu

effective cavity frequency, the molecular analogue of the refractive index of the intracavity medium, and it is the minimal classical mechanism through which the cavity vacuum can bias thermal equilibrium [16]. For liquid CO_2 the same framework predicts strongly modified IR spectra but an unchanged pair structure [16]. Whether water, with its permanent dipole and cooperative hydrogen bonding, behaves differently is the question we address here.

In this work we build up from the single molecule to the liquid in steps that can each be checked. We first validate our simulations against published results for CO_2 and establish the bend and stretch polaritons of a single water molecule (Sec. IV A). We then couple two to four water molecules to one cavity mode to test collective scaling and to find the separations and orientations at which the molecules couple only through the cavity (Sec. IV B). Finally, we simulate liquid water with machine-learned energy, dipole and polarizability models, compute its O–H stretch polaritons, and analyse the hydrogen-bond network and its temperature dependence in direct comparison with experiment (Secs. IV C and IV D).

II. THEORY

III. METHODS

A. Cavity molecular dynamics with `cboamd`

B. Single molecules and small clusters

C. Machine-learned models for liquid water

D. Liquid-water simulations

E. Analysis

IV. RESULTS

A. Validation and single molecules

B. From one to a few molecules

C. Liquid water: O–H stretch polaritons

D. Hydrogen-bond network

V. DISCUSSION

VI. CONCLUSIONS

ACKNOWLEDGMENTS

DATA AVAILABILITY

[1] A. Shalabney, J. George, J. Hutchison, G. Pupillo, C. Genet, and T. W. Ebbesen, Coherent coupling of molecular resonators with a microcavity mode, Nat. Commun. **6**, 5981 (2015).

[2] A. Thomas, J. George, A. Shalabney, M. Dryzhakov, S. J. Varma, J. Moran, T. Chervy, X. Zhong, E. Devaux, C. Genet, J. A. Hutchison, and T. W. Ebbesen, Ground-state chemical reactivity under vibrational coupling to the vacuum electromagnetic field, Angew. Chem. Int. Ed. **55**, 11462 (2016).

[3] A. Thomas, L. Lethuillier-Karl, K. Nagarajan, R. M. A. Vergauwe, J. George, T. Chervy, A. Shalabney, E. Devaux, C. Genet, J. Moran, and T. W. Ebbesen, Tilting a ground-state reactivity landscape by vibrational strong coupling, Science **363**, 615 (2019).

[4] B. Xiang, R. F. Ribeiro, M. Du, L. Chen, Z. Yang, J. Wang, J. Yuen-Zhou, and W. Xiong, Intermolecular vibrational energy transfer enabled by microcavity strong light–matter coupling, Science **368**, 665 (2020).

[5] W. Ahn, J. F. Triana, F. Recabal, F. Herrera, and B. S. Simpkins, Modification of ground-state chemical reactivity via light–matter coherence in infrared cavities, Science **380**, 1165 (2023).

[6] M. V. Imperatore, J. B. Asbury, and N. C. Giebink, Reproducibility of cavity-enhanced chemical reaction rates in the vibrational strong coupling regime, J. Chem. Phys. **154**, 191103 (2021).

[7] G. D. Wiesehan and W. Xiong, Negligible rate enhancement from reported cooperative vibrational strong coupling catalysis, J. Chem. Phys. **155**, 241103 (2021).

[8] T. Fukushima, S. Yoshimitsu, and K. Murakoshi, Inherent promotion of ionic conductivity via collective vibrational strong coupling of water with the vacuum electromagnetic field, J. Am. Chem. Soc. **144**, 12177 (2022).

[9] M. Itatani, N. Iwasa, K. Inaoka, N. Miyasaka, L. Takahashi, K. Takahashi, T. Fukushima, M. Bamba, and K. Murakoshi, Hydrogen-bond network reorganization in liquid water under vibrational strong coupling, arXiv:2606.12508 (2026).

[10] T. E. Li, J. E. Subotnik, and A. Nitzan, Cavity molecular dynamics simulations of liquid water under vibrational ultrastrong coupling, Proc. Natl. Acad. Sci. U.S.A. **117**, 18324 (2020).

[11] A. Z. Lieberherr, S. T. E. Furniss, J. E. Lawrence, and D. E. Manolopoulos, Vibrational strong coupling in liquid water from cavity molecular dynamics, J. Chem. Phys. **158**, 234106 (2023).

[12] J. Bowles, J. De La Fuente Diez, D. Laage, J. Richardi, R. Vuilleumier, and R. Spezia, Liquid water under vibrational strong coupling: An extended cavity Born–Oppenheimer molecular dynamics study, J. Chem. Phys. **163**, 044104 (2025).

[13] T. E. Li, A. Nitzan, and J. E. Subotnik, On the origin of ground-state vacuum-field catalysis: Equilibrium consideration, J. Chem. Phys. **152**, 234107 (2020).

[14] J. Flick, H. Appel, M. Ruggenthaler, and A. Rubio, Cavity Born–Oppenheimer approximation for correlated electron–nuclear-photon systems, J. Chem. Theory Comput. **13**, 1616 (2017).

[15] J. Bonini, I. Ahmadabadi, and J. Flick, Cavity Born–Oppenheimer approximation for molecules and materials via electric field response, J. Chem. Phys. **161**, 154104 (2024).

[16] Y. Li, R. Car, and J. Flick, cboamd: A machine learning molecular dynamics framework for vibrational strong coupling, arXiv:2609.22022 (2026).